Enhancing Data Discovery via Text mining and Harmonization for Fair Data Analysis in Galaxy

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INTRODUCTION

•Harmonization of cohort metadata is essential to performing meaningful cross-cohort analyses, by increasing the interoperability of diverse sets of data. CINECA WP3 teams developed a set of text mining models and workflows to assist a common metadata model by exploring the semantic coding systems of the project cohorts, and designing ontology mapping strategies to produce machine-readable consent ontology (https://www.cineca-project.eu/wp3). Data harmonization tools when combined with a query service such as Beacon can significantly enhance data discovery to assist fair data analysis in Galaxy.

•For example, the new Beacon versions allow filtering on phenotypic data such as gender, age range and diagnosis. However, due to the large number of variables (data items), human curation can hardly scale to the task. This is where automated mapping facilitated by text mining models can help.

In this talk, we outline the steps using data harmonization based on the CINECA developed text mining models to use for fair data analysis in Galaxy. The developed models are based on different approaches such as data repository query, i.e., Zooma, rule-basedrule-based text-mining tool (LexMapr is adopted by FDA to automate the curation of isolation source metadata) and learning-to-rank based, i.e., L2N, to provide standard mappings to free text fields in CINECA cohorts facilitating data discovery.

METHODS

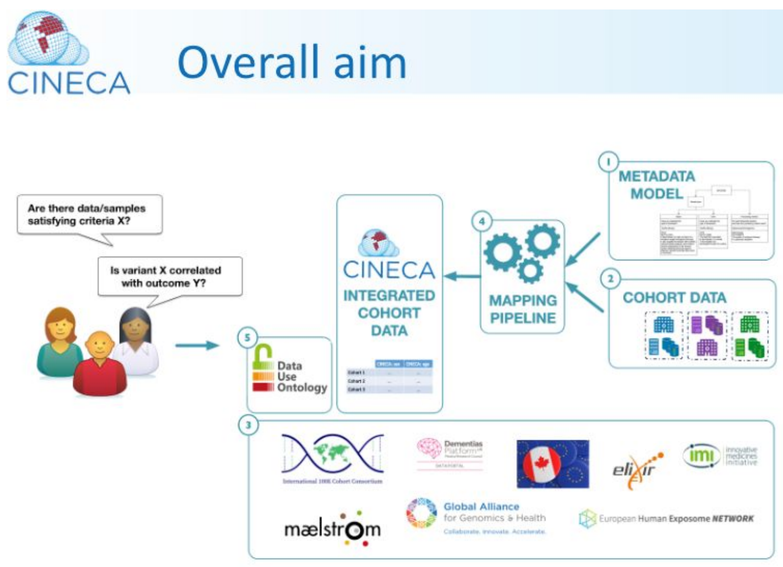

Figure 1. the mapping pipeline (Task 3.4: Automatic generation of metadata from unstructured data set) with an example of the application of the text mining models using GECKO ontology.

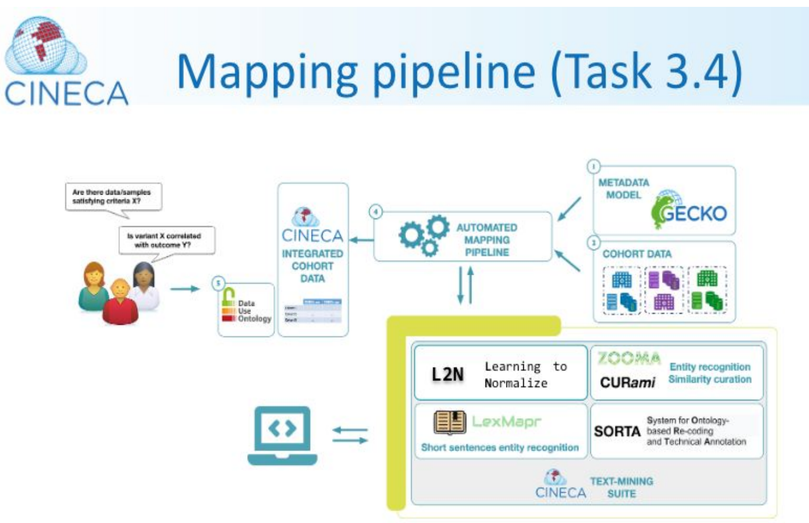

Figure 2: the FAIR data analysis workflow in Galaxy leveraging the mapping pipeline using the text mining models

RESULTS

Figure 4. We have developed an aggregated portal as a concept design to query for annotations using all of the above TM models and is deployed in the EBI Embassy cloud.

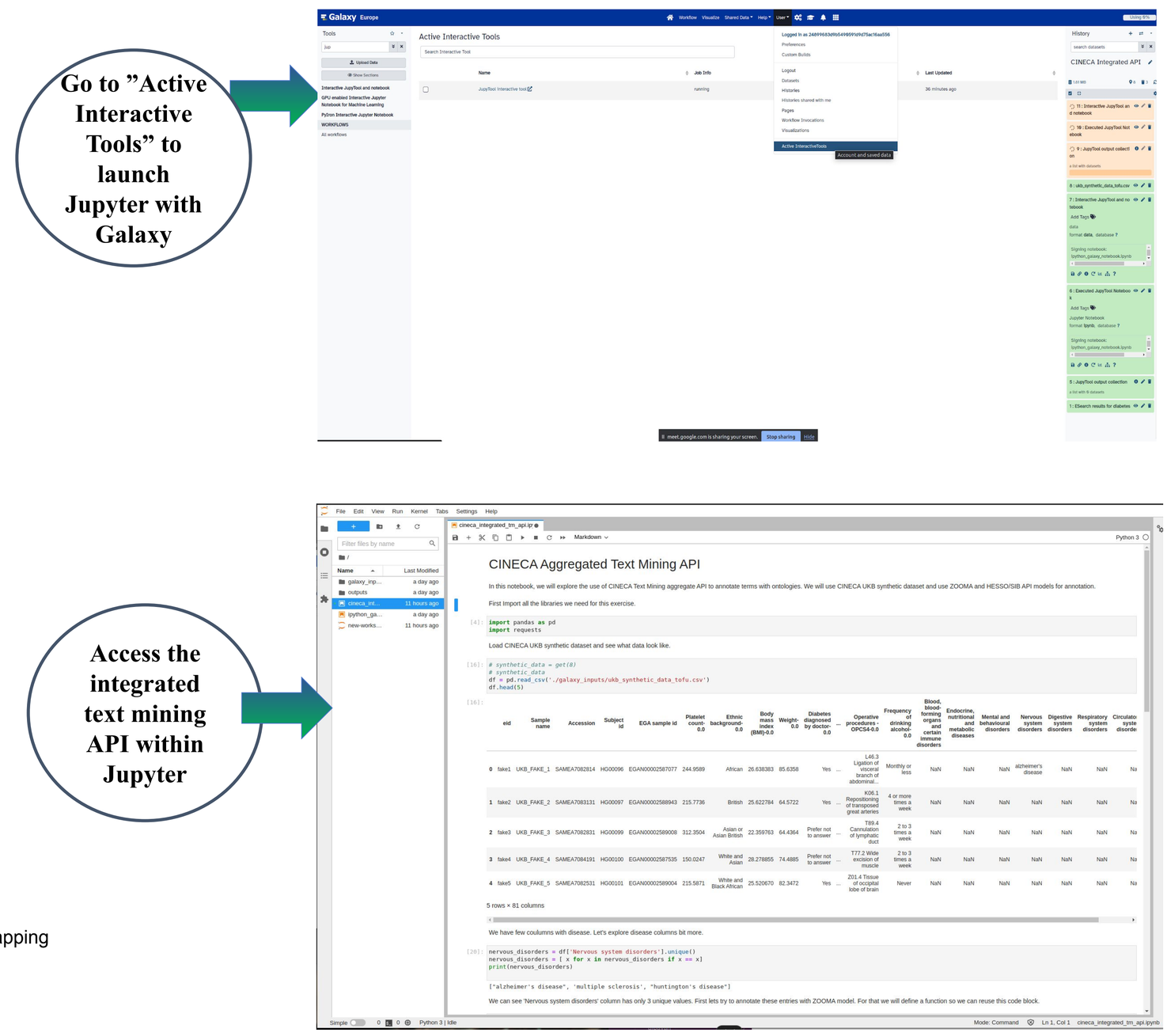

Figure 5. "How to use" in Galaxy showing an example of analysis in Jupyter within Galaxy

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CINECA Common Infrastructure for National Cohorts in Europe, Canada, and Africa

CONCLUSIONS

- In this study, we developed a Galaxy module to execute the steps using data harmonization based on the CINECA developed text mining models to use for fair data analysis in Galaxy. The developed models are based on different approaches such as data repository query, i.e., Zooma, rule-basedrule-based text-mining tool (LexMapr is adopted by FDA to automate the curation of isolation source metadata) and learning-to-rank based, i.e., L2N, to provide standard mappings to free text fields in CINECA cohorts facilitating data discovery.
- Future work will include the integrated text mining API wrapped as a Galaxy tool for direct integration